

HW-01 — Conceptual Modeling

Module M02 | Chapter 3 | Weight 5% | Difficulty ★

Module: M02 Chapter: 3 Weight: 5% Due: End of Week 2

Overview

Build a conceptual model for one of four given real-world systems *before* writing any code. The deliverable is a short structured document that forces you to think clearly about what your simulation will represent and what simplifications you are making.

A strong conceptual model is the foundation of every reliable simulation study.

1 System Choice (choose one)

1. **Drive-through coffee kiosk** — single-lane, one barista; morning peak
2. **University IT help desk** — walk-in and ticket-based requests; two staff
3. **Hospital emergency triage** — two acuity levels; one triage nurse
4. **Grocery checkout** — express lane vs. regular lane; varying demand

Or propose your own system with instructor approval by the end of Week 1.

2 Required Deliverables

2.1 System Description and Scope (3 pts)

Write one paragraph (5–8 sentences) covering:

- The real-world system and its purpose
- The simulation question (what decision or insight motivates the model?)
- The system boundary: what is *in scope* and what is *out of scope*
- The intended level of detail (entity-level vs. aggregate)

2.2 Entity and Resource Tables (4 pts)

Provide two tables:

Entity table — at minimum:

Entity	Key Attributes	Lifetime
(your entities)		

Resource table — at minimum:

Resource	Capacity	Discipline	Notes
(your resources)			

2.3 Event List and State Variables (4 pts)

List all event types (arrival, service start, service end, etc.) and all state variables your simulation engine would maintain. For each state variable give its type (integer, real, boolean) and range.

2.4 Performance Measures (3 pts)

List at least 3 performance measures with units that your model will produce. Examples: mean wait in queue (minutes), server utilization (dimensionless), throughput (customers/hour).

2.5 Assumptions (3 pts)

State at least 6 assumptions. For each, write *one sentence of justification*. Assumptions without justification receive no credit.

2.6 Flow Diagram (3 pts)

Draw a flow diagram showing how entities move through the system. At minimum it must show: arrivals, queueing, service, and departure. Branching logic (e.g., reneging, routing) must be shown. Hand-drawn and photographed is acceptable; digital is preferred.

3 Format and Submission

- Length: 2–4 pages (excluding flow diagram)
- Submit as PDF via GitHub Classroom
- File name: hw01_yournetid.pdf

Grading Summary

Component	Points
System description and scope	3
Entity and resource tables	4
Event list and state variables	4
Performance measures (with units)	3
Assumptions (≥ 6 , each justified)	3
Flow diagram (clear, labeled)	3
Total	20

Full rubric: [rubrics/homework_rubric.pdf](#) | Solutions: the instructor package (available to instructors on request).